Using large-scale seismometer data to detect fin whale calls in the North Atlantic: implications for population density estimation

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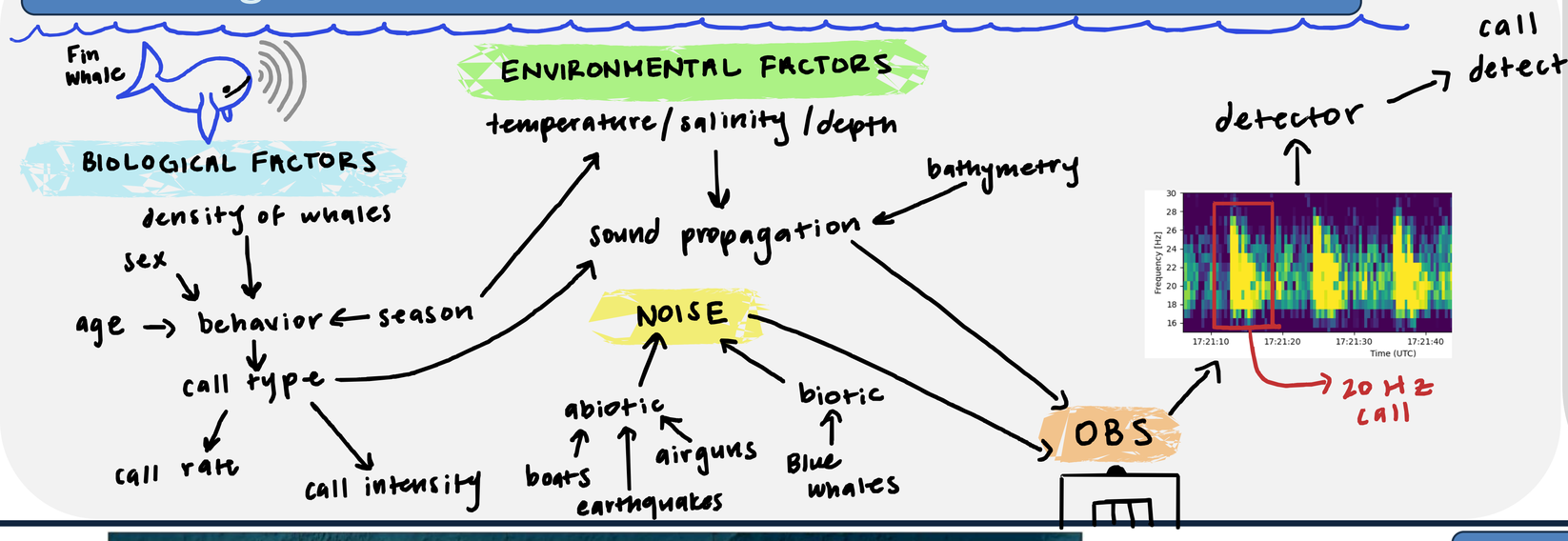
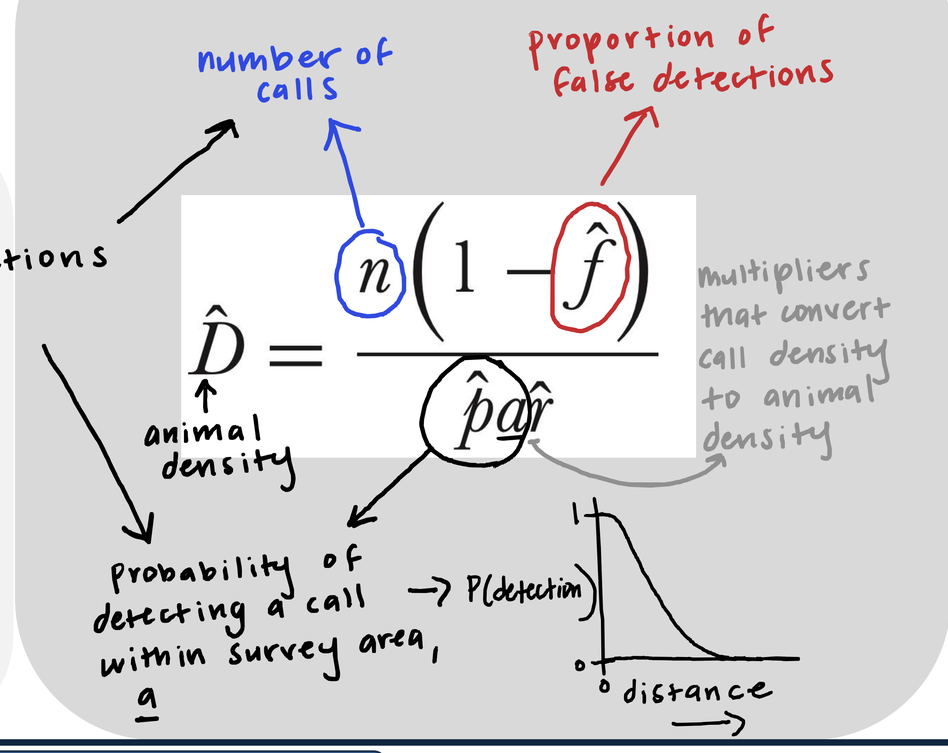
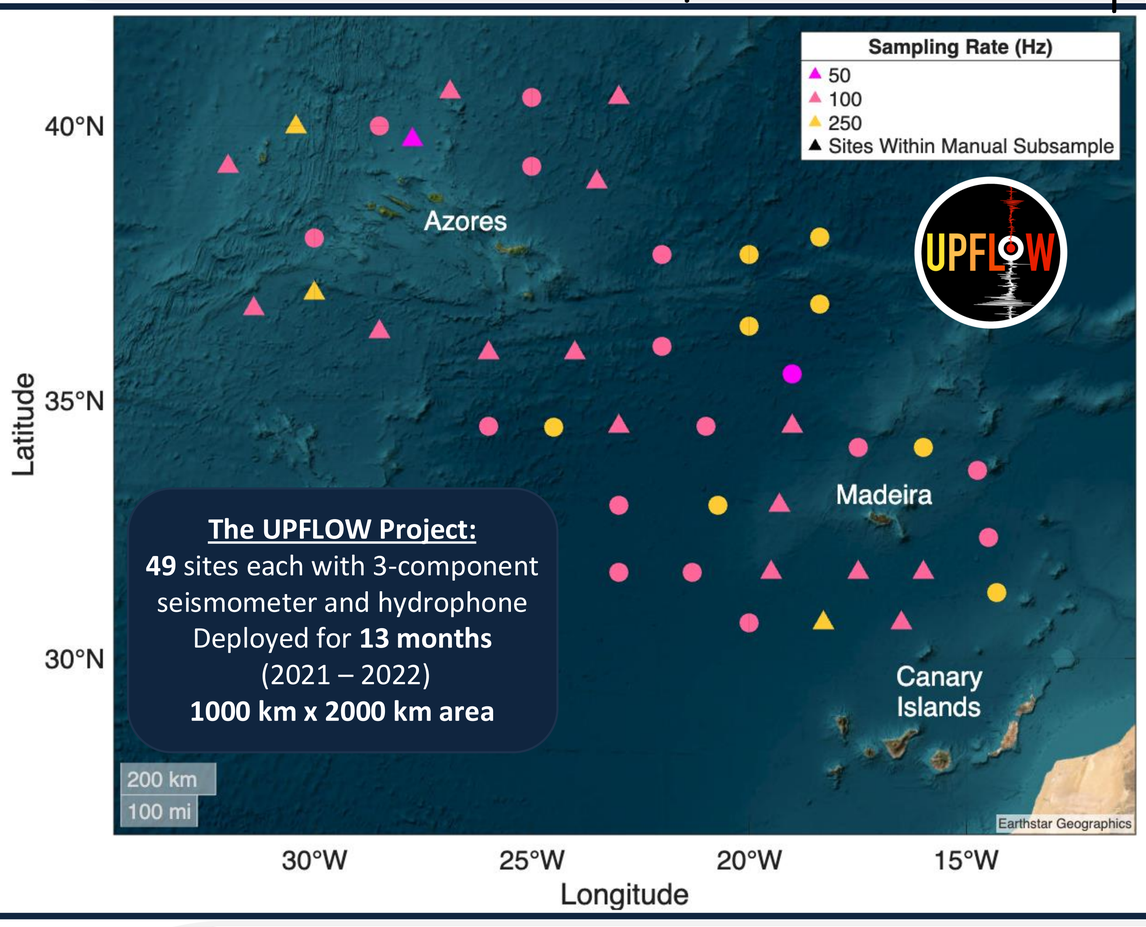
Background: Factors that Influence Call Detection



Background: Density Estimation (DE)

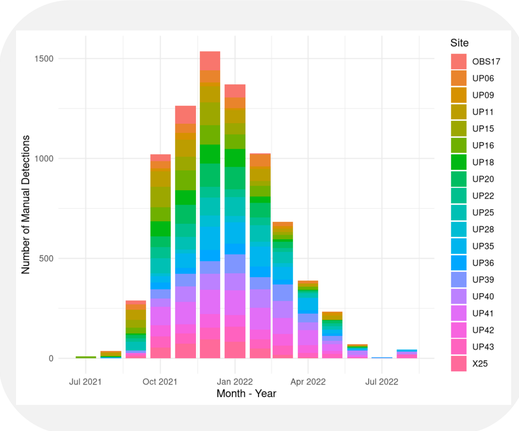




Methods

Manual Subsample

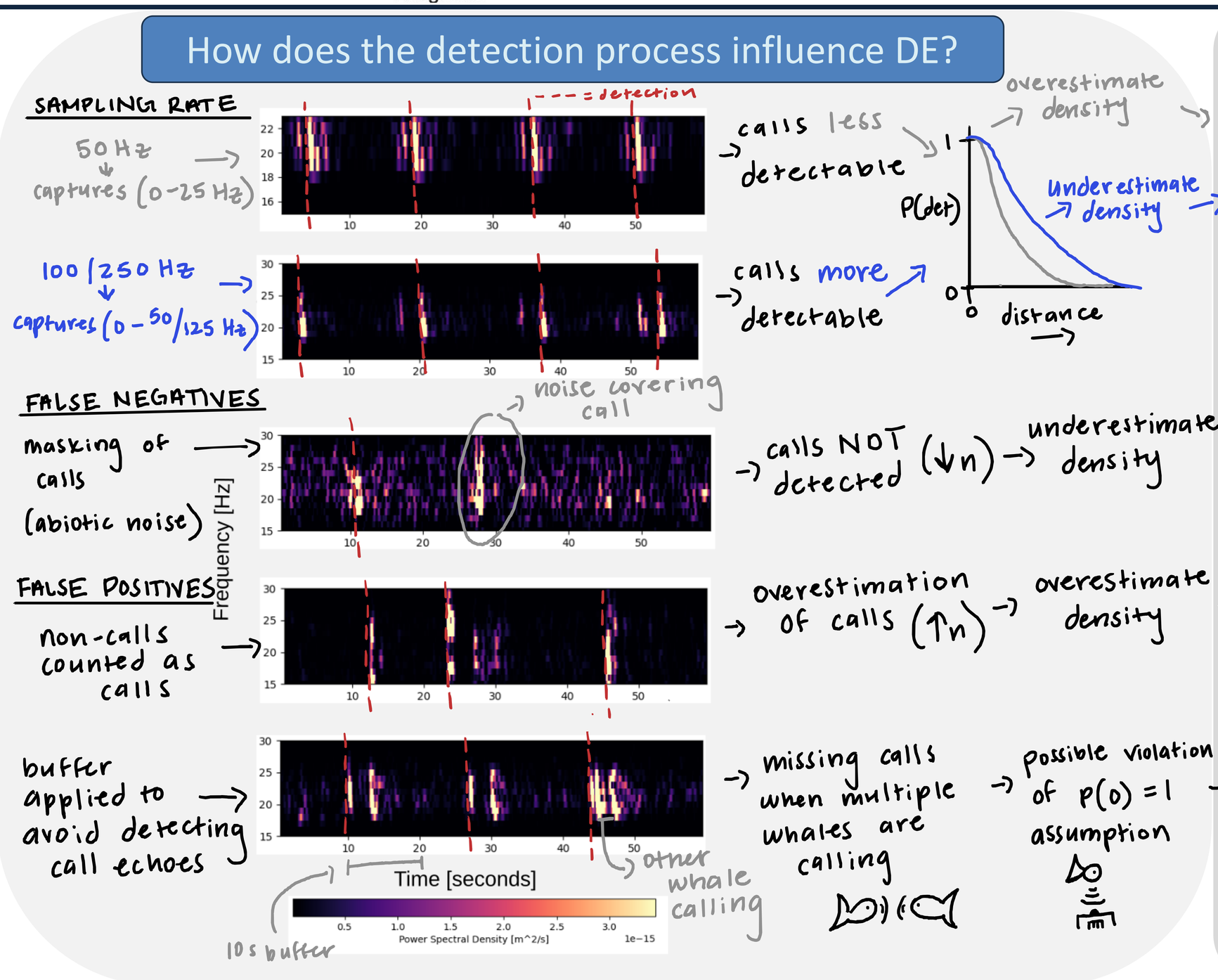
Spatially balanced sample of 19 sites → systematic random sample of 1-minute periods → manual detection of calls →



Automatic Detection

use manual detections to investigate temporal and spatial variability → apply automatic detector (spectrogram cross correlation) → evaluate detector

How does the detection process influence DE?



possible violation of p(0) = 1 assumption → account for proportion of calls missed

Solutions

one detector per sampling rate (50 Hz + 100/250 Hz) → $\hat{f}$ that reflects site + temporal variability → accounted for in $\hat{p}$

CONCLUSION: When using large datasets, site and temporal variability of detections must be accounted for to avoid biased density estimates

References & Appendix:

Funded by: